

Question Number	Answer	Additional guidance	Mark
3(a)	An explanation that makes reference to three of the following points: • many hydrogen bonds form between {cellulose molecules / microfibrils} (1) • microfibrils arranged in {layers / sheets} at different angles (1) • (microfibril arrangement / it) gives added {tensile strength / rigidity / structural support / stability} (to maintain the structure of the xylem) (1) • gaps between cellulose microfibrils allow movement of {water / mineral ions / dissolved solutes} (1)	ACCEPT microfibrils arranged in {criss-cross pattern / a mesh / matrix / different directions} IGNORE {lignin / secondary thickening} providing strength ACCEPT {embedded in / held together by} {calcium pectate / hemicellulose} to provide strength REJECT if in context of {pits/plasmodesmata} and not cell wall	(3)

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3(b)(i)	An answer that makes reference to two of the following points: • {in normal conditions / with no treatment} A and B absorb similar mass of calcium ions (1) • both chemicals reduce the {mean mass / absorption} of calcium ions (1) • EDTA {reduces mass in type A more / has greater effect on type A / reduces absorption more in type A} (1) • citric acid {reduces mass in type B more / has greater effect on type B / reduces absorption more in type B} (1)	needs to be about effect of treatment clear statement required IGNORE no treatment had higher mass of calcium ions IGNORE both treatments improve the transport of calcium ions from root to other parts of plant ACCEPT converse IGNORE A has more calcium ions with EDTA than type B / converse ACCEPT converse IGNORE B has more calcium ions with citric acid than A / converse	(2)

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3(b)(ii)	An answer that makes reference to two of the following points: • fewer calcium ions (1) - resulting in reduced {calcium pectate / middle lamella} (1) • therefore, EDTA reduces the {strength / stability} of the cell wall in the roots of type A (1)	REJECT calcium unqualified IGNORE calcium pectate strengthens cell wall	(2)